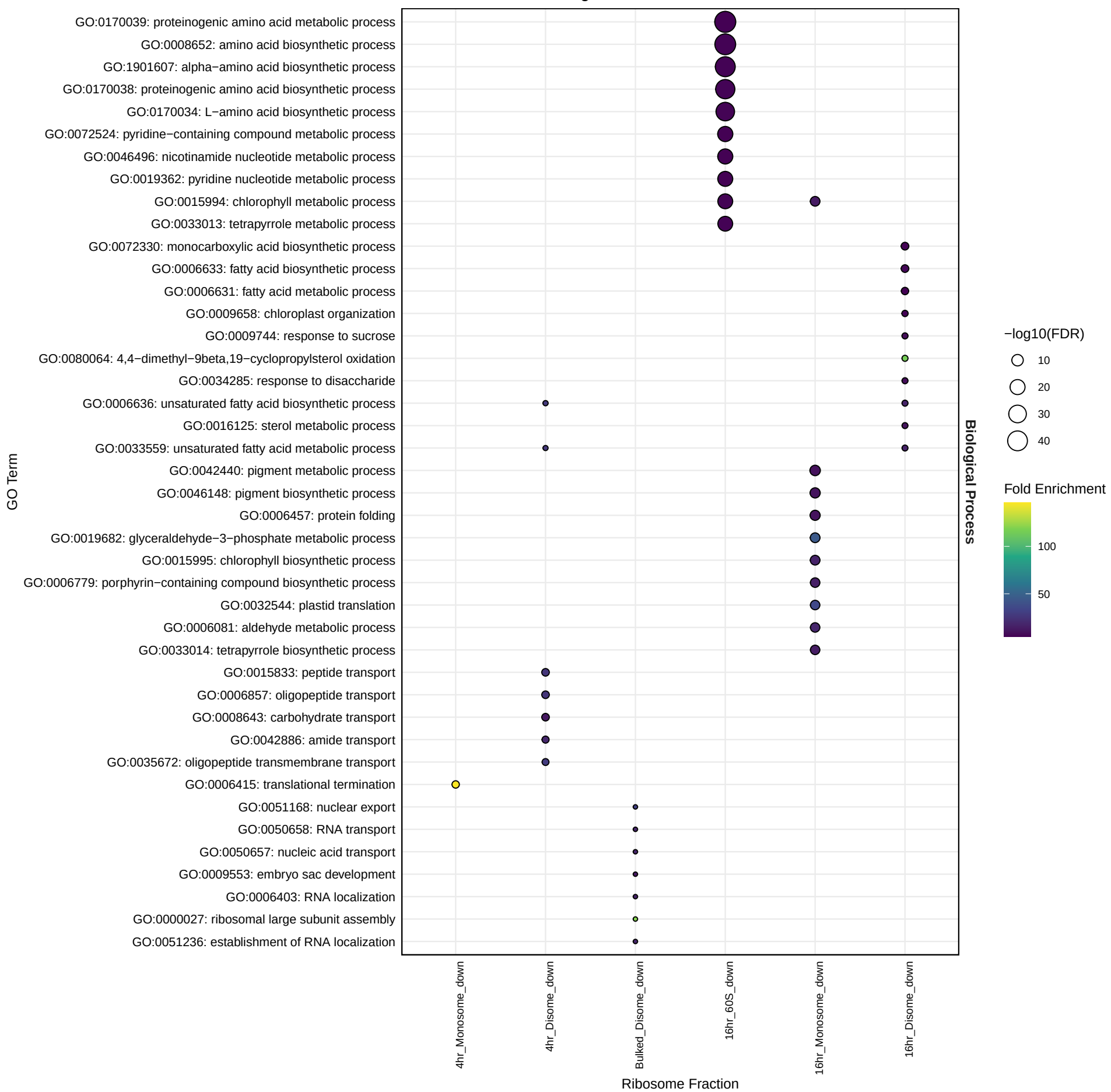
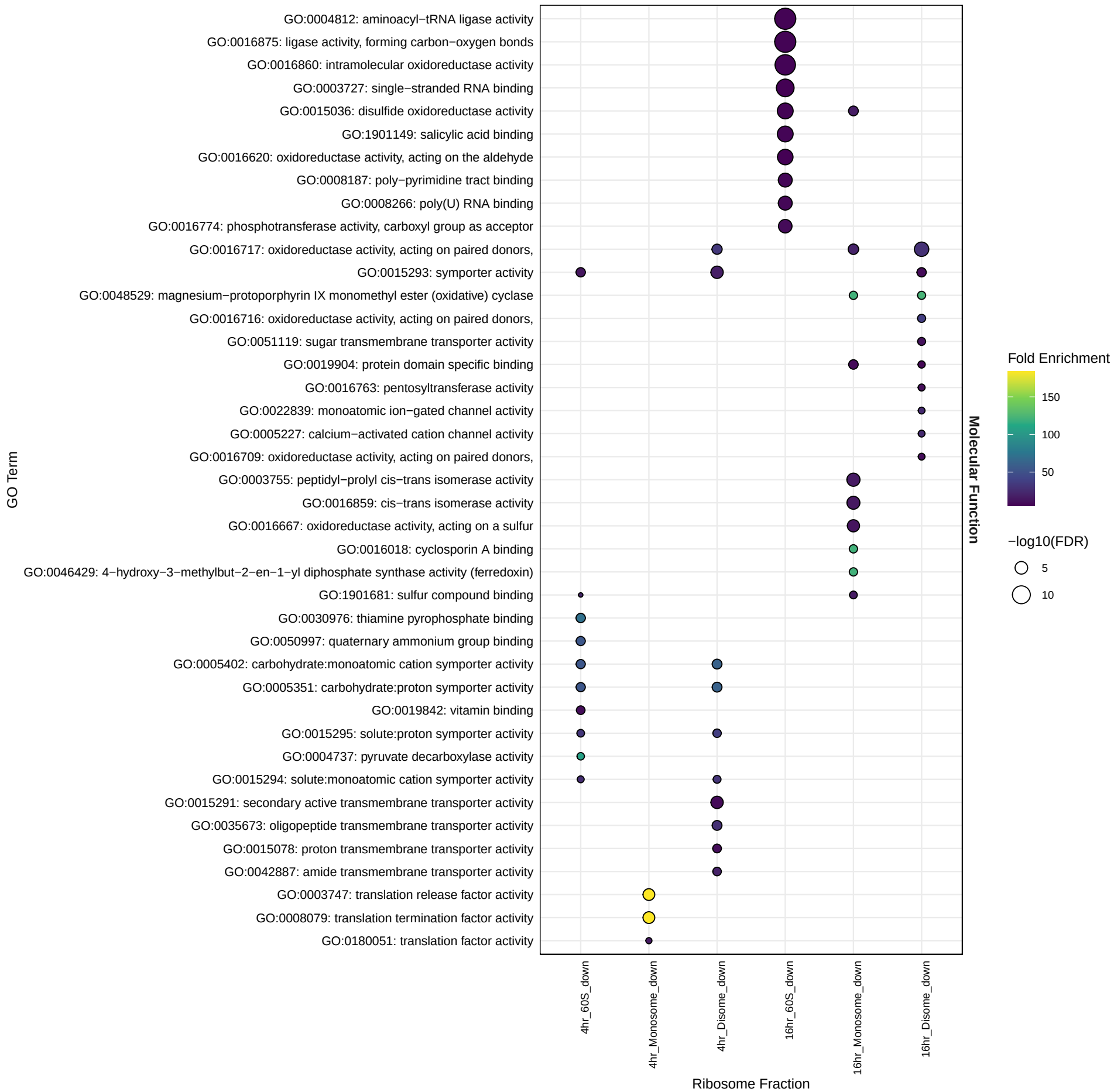


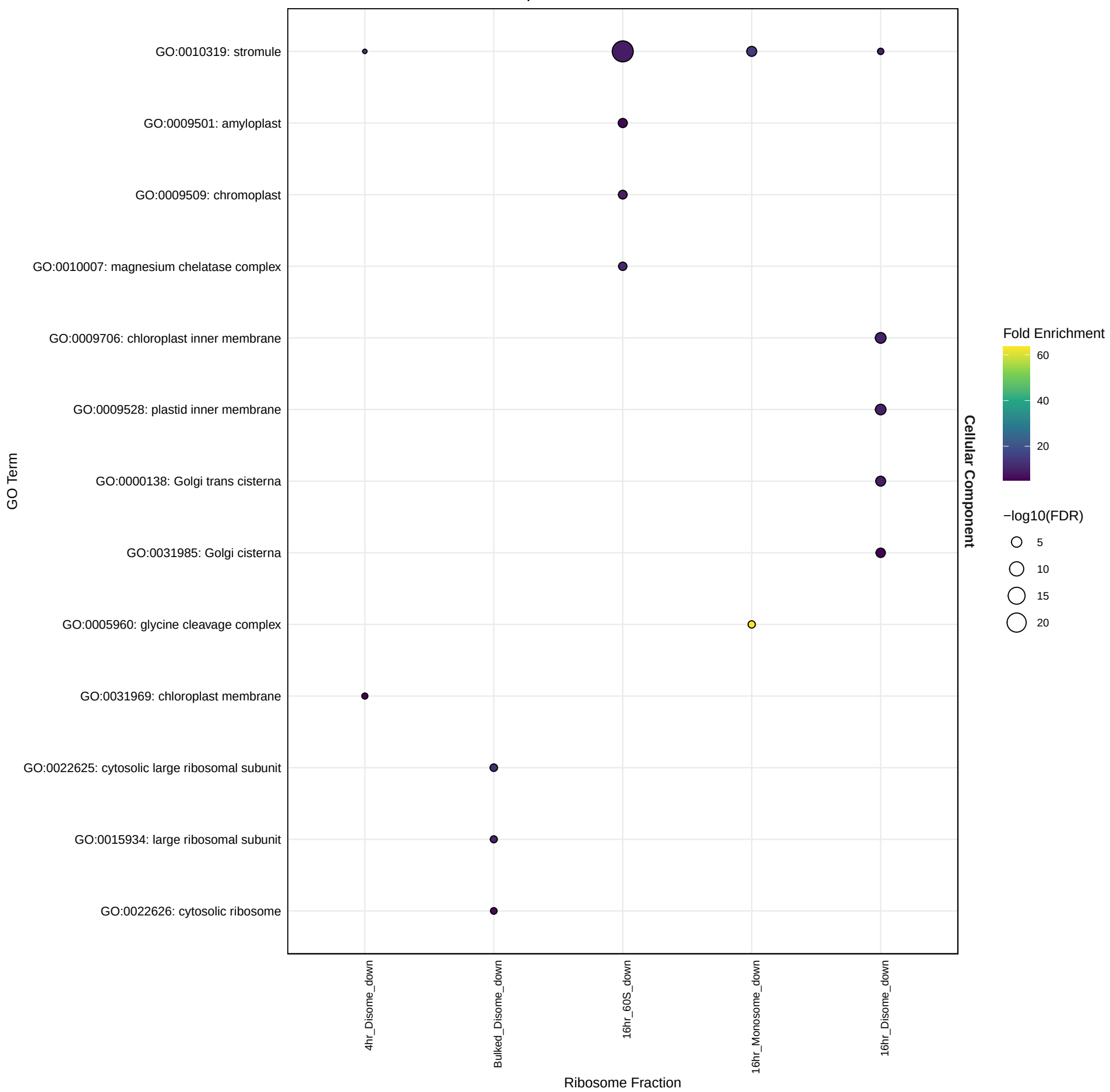
DOWN Clusters – Biological Process



DOWN Clusters – Molecular Function



DOWN Clusters – Cellular Component



DOWN Clusters – All Categories

GO Term

GO:0170039: proteinogenic amino acid metabolic process
 GO:0008652: amino acid biosynthetic process
 GO:1901607: alpha-amino acid biosynthetic process
 GO:0170038: proteinogenic amino acid biosynthetic process
 GO:0170034: L-amino acid biosynthetic process
 GO:0072524: pyridine-containing compound metabolic process
 GO:0046496: nicotinamide nucleotide metabolic process
 GO:0019362: pyridine nucleotide metabolic process
 GO:0015994: chlorophyll metabolic process
 GO:0033013: tetrapyrrole metabolic process
 GO:0072330: monocarboxylic acid biosynthetic process
 GO:0006633: fatty acid biosynthetic process
 GO:0006631: fatty acid metabolic process
 GO:0009658: chloroplast organization
 GO:0009744: response to sucrose
 GO:0080064: 4,4-dimethyl-9beta,19-cyclopropylsterol oxidation
 GO:0034285: response to disaccharide
 GO:0006636: unsaturated fatty acid biosynthetic process
 GO:0016125: sterol metabolic process
 GO:0033559: unsaturated fatty acid metabolic process
 GO:0042440: pigment metabolic process
 GO:0046148: pigment biosynthetic process
 GO:0006457: protein folding
 GO:0019682: glyceraldehyde-3-phosphate metabolic process
 GO:0015995: chlorophyll biosynthetic process
 GO:0006779: porphyrin-containing compound biosynthetic process
 GO:0032544: plastid translation
 GO:0006081: aldehyde metabolic process
 GO:0033014: tetrapyrrole biosynthetic process
 GO:0015833: peptide transport
 GO:0006857: oligopeptide transport
 GO:0008643: carbohydrate transport
 GO:0042886: amide transport
 GO:0035672: oligopeptide transmembrane transporter activity
 GO:0006415: translational termination
 GO:0051168: nuclear export
 GO:0050658: RNA transport
 GO:0050657: nucleic acid transport
 GO:0009553: embryo sac development
 GO:0006403: RNA localization
 GO:0000027: ribosomal large subunit assembly
 GO:0051236: establishment of RNA localization

GO:0004812: aminoacyl-tRNA ligase activity
 GO:0016875: ligase activity, forming carbon-oxygen bonds
 GO:0016860: intramolecular oxidoreductase activity
 GO:0003727: single-stranded RNA binding
 GO:0015036: disulfide oxidoreductase activity
 GO:1901149: salicylic acid binding
 GO:0016620: oxidoreductase activity, acting on the aldehyde
 GO:0008187: poly-pyrimidine tract binding
 GO:0008266: poly(U) RNA binding
 GO:0016774: phosphotransferase activity, carboxyl group as acceptor
 GO:0016717: oxidoreductase activity, acting on paired donors,
 GO:0048529: magnesium-protoporphyrin IX monomethyl ester (oxidative) cyclase
 GO:0016716: oxidoreductase activity, acting on paired donors,
 GO:0051119: sugar transmembrane transporter activity
 GO:0019904: protein domain specific binding
 GO:0016763: pentosyltransferase activity
 GO:0022839: monoatomic ion-gated channel activity
 GO:0005227: calcium-activated cation channel activity
 GO:0016709: oxidoreductase activity, acting on paired donors,
 GO:0003755: peptidyl-prolyl cis-trans isomerase activity
 GO:0016859: cis-trans isomerase activity
 GO:0016667: oxidoreductase activity, acting on a sulfur
 GO:0016018: cyclosporin A binding
 GO:0046429: 4-hydroxy-3-methylbut-2-en-1-yl diphosphate synthase activity (ferredoxin)
 GO:1901681: sulfur compound binding
 GO:0030976: thiamine pyrophosphate binding
 GO:0050997: quaternary ammonium group binding
 GO:0005402: carbohydrate:monoatomic cation symporter activity
 GO:0005351: carbohydrate:proton symporter activity
 GO:0019842: vitamin binding
 GO:0015295: solute:proton symporter activity
 GO:0004737: pyruvate decarboxylase activity
 GO:0015294: solute:monoatomic cation symporter activity
 GO:0015291: secondary active transmembrane transporter activity
 GO:0035673: oligopeptide transmembrane transporter activity
 GO:0015078: proton transmembrane transporter activity
 GO:0042887: amide transmembrane transporter activity
 GO:0003747: translation release factor activity
 GO:0008079: translation termination factor activity
 GO:0180051: translation factor activity

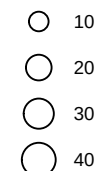
GO:0010319: stromule
 GO:0009501: amyloplast
 GO:0009509: chromoplast
 GO:0010007: magnesium chelatase complex
 GO:0009706: chloroplast inner membrane
 GO:0009528: plastid inner membrane
 GO:0000138: Golgi trans cisterna
 GO:0031985: Golgi cisterna
 GO:0005960: glycine cleavage complex
 GO:0031969: chloroplast membrane
 GO:0022625: cytosolic large ribosomal subunit
 GO:0015934: large ribosomal subunit
 GO:0022626: cytosolic ribosome

Biological Process

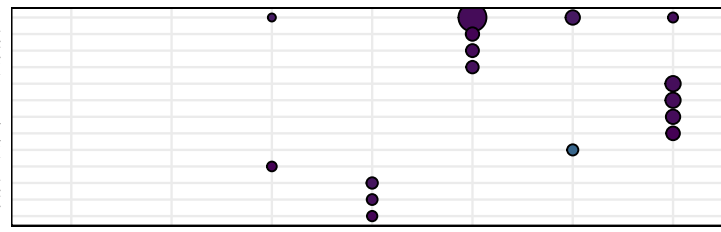
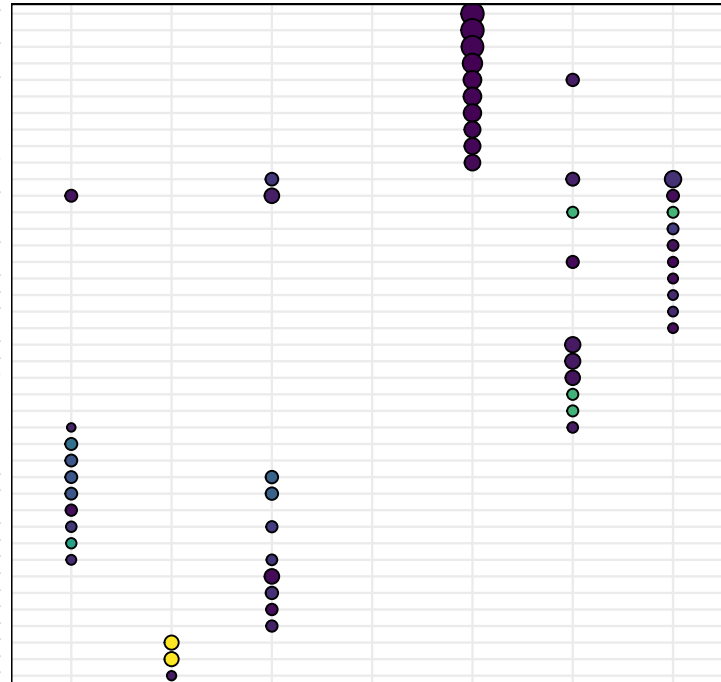
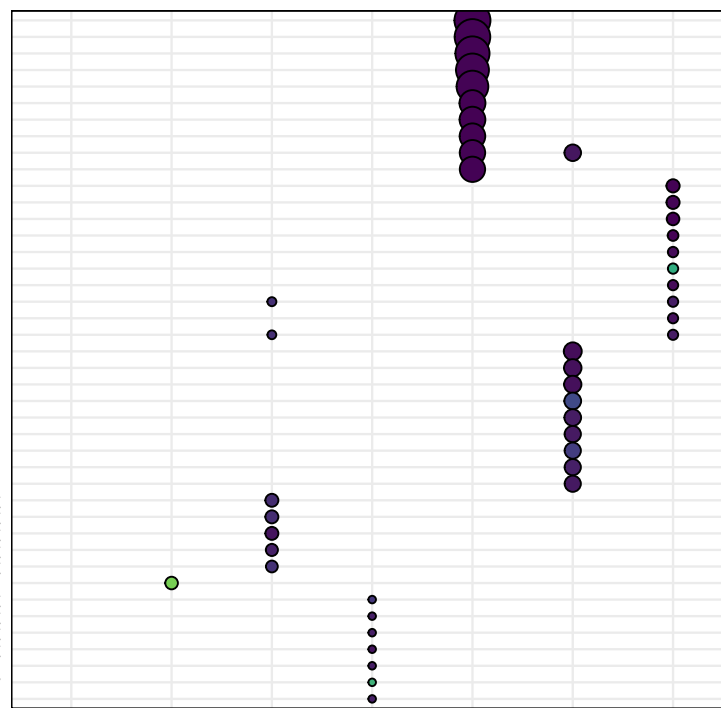
Molecular Function

Cellular Component

$-\log_{10}(\text{FDR})$



Fold Enrichment



4hr_60S_down 4hr_Monosome_down 4hr_Disome_down Bulked_Disome_down 16hr_60S_down 16hr_Monosome_down 16hr_Disome_down

Ribosome Fraction